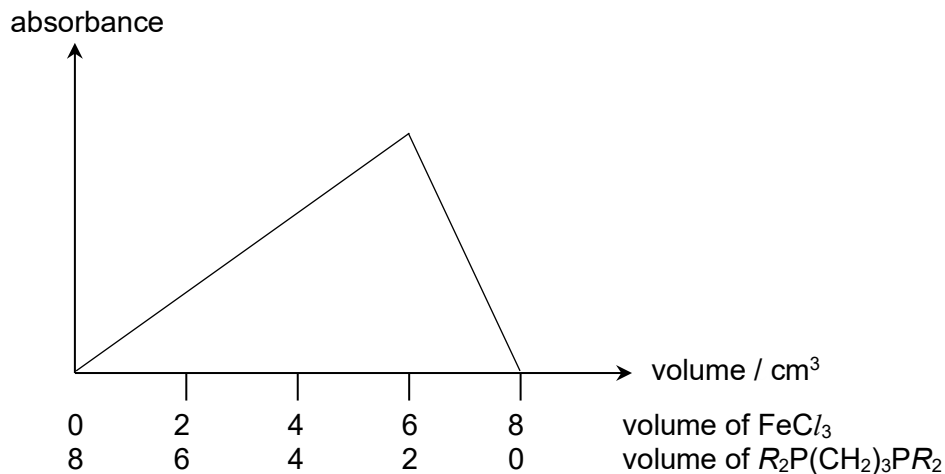


29 In this question, R' represents a phenyl group.

2-bis(diphenylphosphino)propane, $R_2P(CH_2)_3PR_2$, is a commonly used ligand which forms a complex with many metal ions.

In the graph below, the intensity of visible light absorbance for different mixtures containing $1.00 \times 10^{-3} \text{ mol dm}^{-3} \text{ FeCl}_3(\text{aq})$ and $9.00 \times 10^{-3} \text{ mol dm}^{-3} R_2P(CH_2)_3PR_2$ are shown.



Which statement is incorrect?

- A** When excess aqueous silver nitrate is added to the mixture with the highest absorbance, $1.8 \times 10^{-5} \text{ mol}$ of AgCl is formed.
- B** H_2O molecule is a weaker ligand than $R_2P(CH_2)_3PR_2$.
- C** The coordination number of the complex formed is 6.
- D** The overall charge of the complex formed is 0.